

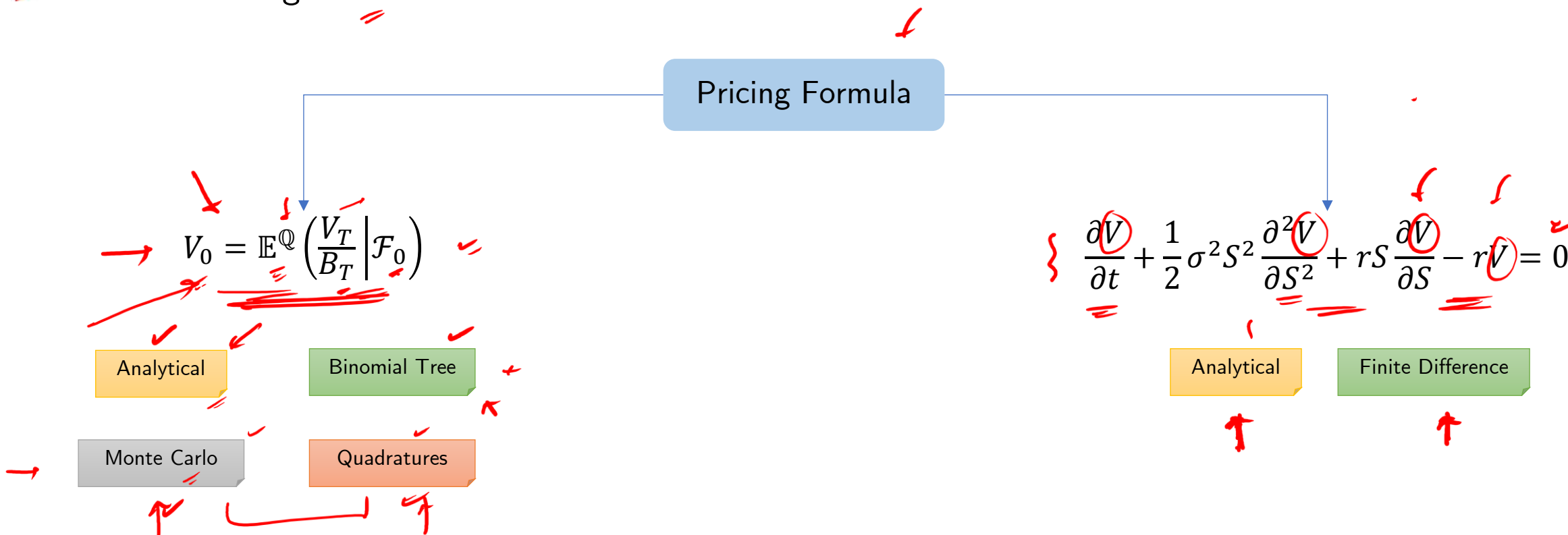
The Black-Scholes Model

(1. European Call Option Pricing)

Option Pricing Formula

Consider an equity derivative $V(S, t)$. There are two main pricing approaches

- Price via **Expectation**
- Price via solving a **PDE**



European Call Option (Analytical)

Consider a European call option having a payoff $V_T = (S_T - K)^+$, $\max(S_T - K, 0)$

Let's calculate the analytical price of this call option at $t = 0$

$$\begin{aligned}
 V_0 &= \mathbb{E}^{\mathbb{Q}}\left(\frac{V_T}{B_T} \middle| \mathcal{F}_0\right) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] \\
 &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K) 1_{S_T > K}] \\
 &= e^{-rT} \mathbb{E}^{\mathbb{Q}}[S_T 1_{S_T > K}] - K e^{-rT} \mathbb{E}^{\mathbb{Q}}[1_{S_T > K}]
 \end{aligned}$$

$B_T = e^{rT}$
 $(S_T - K)^+ \rightarrow S_T - K, S_T > K$
 $(S_T - K)^+ \rightarrow 0, S_T < K$
 $1_{S_T > K} \rightarrow 1, S_T > K$
 $1_{S_T > K} \rightarrow 0, S_T < K$

(II) $\underbrace{e^{-rT} \mathbb{E}^{\mathbb{Q}}[S_T 1_{S_T > K}]}$
 (I) $\underbrace{K e^{-rT} \mathbb{E}^{\mathbb{Q}}[1_{S_T > K}]}$

$$\begin{aligned}
 \mathbb{E}[1_A] &= P(A) \\
 &= 1 \times P(A) + 0 \times (1 - P(A)) = P(A)
 \end{aligned}$$

European Call Option (Analytical)

Let's evaluate (I)

$$Ke^{-rT} \mathbb{Q}(1_{S_T > K}) = Ke^{-rT} \mathbb{Q}\left(S_0 \exp\left\{\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\tilde{W}_T\right\} > K\right)$$

$$= Ke^{-rT} \mathbb{Q}\left(\exp\left\{\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\tilde{W}_T\right\} > \frac{K}{S_0}\right)$$

$$= Ke^{-rT} \mathbb{Q}\left(\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\tilde{W}_T > \ln\left(\frac{K}{S_0}\right)\right)$$

$$= Ke^{-rT} \mathbb{Q}\left(\tilde{W}_T > \frac{\ln\left(\frac{K}{S_0}\right) - \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma}\right)$$

$$= Ke^{-rT} \mathbb{Q}\left(Z < \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right)$$

$$= Ke^{-rT} N(d_2)$$

S_T

$\tilde{W}_T$ is a $\mathbb{Q}$ -Brownian

$$a > b \Rightarrow \ln a > \ln b$$

$$P(Z < a) = P(Z > -a)$$

$$P(Z < a) = \text{cdf}(a)$$

$$\mathbb{Q}\left(\frac{\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\tilde{W}_T}{e} > \ln\left(\frac{K}{S_0}\right)\right)$$

$$\mathbb{Q}\left(\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\tilde{W}_T > \ln\left(\frac{K}{S_0}\right)\right)$$

$$d_2 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$$



$$W_T = (\sqrt{T}) \cdot Z$$

European Call Option (Analytical)

Now Let's evaluate (II)

$$\begin{aligned}
 e^{-rT} \mathbb{E}^{\mathbb{Q}}[S_T 1_{S_T > K}] &= e^{-rT} \mathbb{E}^{\mathbb{Q}} \left[\left(S_0 \exp \left\{ \left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \tilde{W}_T \right\} \right) \times 1_{S_T > K} \right] \\
 &= \mathbb{E}^{\mathbb{Q}} \left[\left(S_0 \exp \left\{ \left(-\frac{1}{2} \sigma^2 \right) T + \sigma \tilde{W}_T \right\} \right) \times 1_{S_T > K} \right] \\
 &= \mathbb{E}^{\mathbb{Q}^*} [(S_0) \times 1_{S_T > K}] = \mathbb{E}^{\mathbb{Q}^*} \left(S_0 \cdot 1_{S_T > K} \cdot \frac{d\mathbb{Q}^*}{d\mathbb{Q}} \right)
 \end{aligned}$$

Handwritten notes: $e^{-\frac{1}{2}\sigma^2 T + \sigma \tilde{W}_T}$ (circled and underlined)

$$\begin{aligned}
 \mathbb{E}^{\mathbb{Q}^*}(\cdot) &= \mathbb{E}^{\mathbb{Q}} \left(\cdot \times \frac{d\mathbb{Q}^*}{d\mathbb{Q}} \right) \\
 \int (\cdot) d\mathbb{Q}^* &= \int (\cdot) \frac{d\mathbb{Q}^*}{d\mathbb{Q}} d\mathbb{Q} \\
 \frac{d\mathbb{Q}^*}{d\mathbb{Q}} &= e^{-\frac{1}{2}\sigma^2 T + \sigma \tilde{W}_T} \quad (\text{RN Derivative}) \\
 W_T^{\mathbb{Q}^*} &= \tilde{W}_T - \sigma T \quad (\text{C.M.G}) \\
 W_T^{\text{new}} &= W_T^{\text{old}} + \lambda T
 \end{aligned}$$

Handwritten notes: $e^{-\frac{1}{2}\sigma^2 T + \sigma \tilde{W}_T}$ (circled), $W_T^{\text{new}} = W_T^{\text{old}} + \lambda T$ (circled)

The dynamics of S_T under stock-measure $\mathbb{Q}^*$ is given by

$$\begin{aligned}
 S_T &= S_0 \exp \left\{ \left(r - \frac{1}{2} \sigma^2 \right) T + \sigma (W_T^{\mathbb{Q}^*} + \sigma T) \right\} \\
 &= S_0 \exp \left\{ \left(r + \frac{1}{2} \sigma^2 \right) T + \sigma W_T^{\mathbb{Q}^*} \right\}
 \end{aligned}$$

Handwritten notes: $(r + \frac{1}{2}\sigma^2)$ (circled), $W_T^{\mathbb{Q}^*}$ (circled)

$$\begin{aligned}
 \mathbb{E}^{\mathbb{Q}^*}(1_{S_T > K}) &= \mathbb{E}^{\mathbb{Q}} \left(1_{S_T > K} \cdot \frac{d\mathbb{Q}^*}{d\mathbb{Q}} \right) \\
 &= \mathbb{E}^{\mathbb{Q}} \left(1_{S_T > K} \cdot e^{-\frac{1}{2}\sigma^2 T + \sigma \tilde{W}_T} \right) \\
 &= \mathbb{E}^{\mathbb{Q}} \left(1_{S_T > K} \cdot e^{-\frac{1}{2}\sigma^2 T + \sigma \tilde{W}_T} \right) \\
 &= \mathbb{E}^{\mathbb{Q}} \left(1_{S_T > K} \cdot e^{-\frac{1}{2}\sigma^2 T + \sigma \tilde{W}_T} \right)
 \end{aligned}$$

Handwritten notes: $(r + \sigma^2) - \frac{1}{2}\sigma^2 = (+\frac{1}{2}\sigma^2)$ (circled)

Note : The drift of S_t is $(r + \sigma^2)$ instead of r under the stock measure

$$\begin{array}{ccc}
 \mathbb{P} & \mathbb{Q} & \mathbb{Q}^* \\
 \mu & r & (r + \sigma^2)
 \end{array}$$

European Call Option (Analytical)

This leads to (II)

$$= \mathbb{E}^{\mathbb{Q}^*} [S_0 \times 1_{S_T > K}]$$

$$= S_0 \mathbb{E}^{\mathbb{Q}^*} [1_{S_T > K}]$$

$$W_T^* \rightarrow \bar{W}_T \quad (FAG)$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Replace r by $(r + \sigma^2)$ in the previous derivation

$$= S_0 N(d_1) \quad d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$$

Price of the European Call Option

$$V_0 = S_0 N\left(\frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right) - Ke^{-rT} N\left(\frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right)$$